

Classification of Living Beings

A) Choose the best alternative.

1) The microorganisms that are harmful and cause diseases are called

- (i) Pathogens (ii) microbes (iii) autotrophs (iv) none of these

2) Virus was first isolated by.

- (i) Robert Ennenberg (ii) Louis Pasteur
(iii) Iwanowski (iv) Cohn

3) Virus can infect.

- (i) animal (ii) plant
(iii) bacteria (iv) ~~all of them~~

4) A bacterial cell does not contain

- (i) cell wall (ii) cytoplasm
(iii) chlorophyll (iv) ~~well developed nucleus~~

5) Microorganism is used in the preparation of bread

- (i) virus (ii) algae
(iii) bacteria (iv) ~~yeast~~

6) Which of these is not a fungus

- (i) yeast (ii) mushroom
(iii) mould (iv) bacillus

B) Very short answer questions

1) What is classification of living beings?

→ The system of grouping of living organisms on the basis of similarities and dissimilarities is called classification of living organisms.

2) What is the basic unit of classification?
→ Species is the basic unit of classification.

3) What are microorganisms?
→ Microorganisms are very small organisms that cannot be seen with the naked eye and can be seen with a ~~micro~~ microscope.

4) What is Fermentation?
→ Fermentation is the process in which microorganisms convert sugar into alcohol, acids or gases in the absence of oxygen.

5) What is virus?
→ A virus is a non-cellular infectious particle that reproduces only inside a living host cell.

6) Name any two viruses that cause human diseases.

-
- Influenza virus
 - HIV

7) Name any two useful bacteria.

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- Lactobacillus
 - Rhizobium

8) Name any two diseases caused by amoeba.

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- Amoeba Amoebic dysentery
 - Amoebiasis

Q Write the nature of fungi.

→ Fungi are heterotrophic organisms and mostly obtain food from dead and decaying matter.

10) Name any two microorganisms studied by you.

-
- Bacteria
 - Yeast

11) What is preservation of food?

→ Food preservation is the ~~food~~ process of preventing food from spoiling and increasing its shelf life.

Q Short answer type questions

1) Write any two importance of classification of living beings.

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- It makes the study of a large number of organisms easier.
 - It helps to identify and compare organisms easily.

2) Write any two drawbacks of two-kingdom classification.

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- It does not clearly distinguish between prokaryotes and eukaryotes.
 - It places unicellular and multicellular organisms in the same group.

3) Write any two advantages of five-kingdom classification.

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- It separates prokaryotic organisms from eukaryotic organisms.
 - It separates unicellular organisms from multicellular organisms.

4) Write any two characteristics of amoeba.
→ • Amoeba is a unicellular organism.
• It moves and captures food with the help of pseudopodia.

5) Write any two useful effects of bacteria.
→ • Bacteria help in making curd and cheese.
• Some bacteria increase soil fertility by fixing nitrogen.

6) Write any two harmful effects of fungi.
→ • Fungi cause diseases in humans, animals and plants.
• Fungi spoil food and other stored materials.

7) Draw a structure of bacteriophage.
→

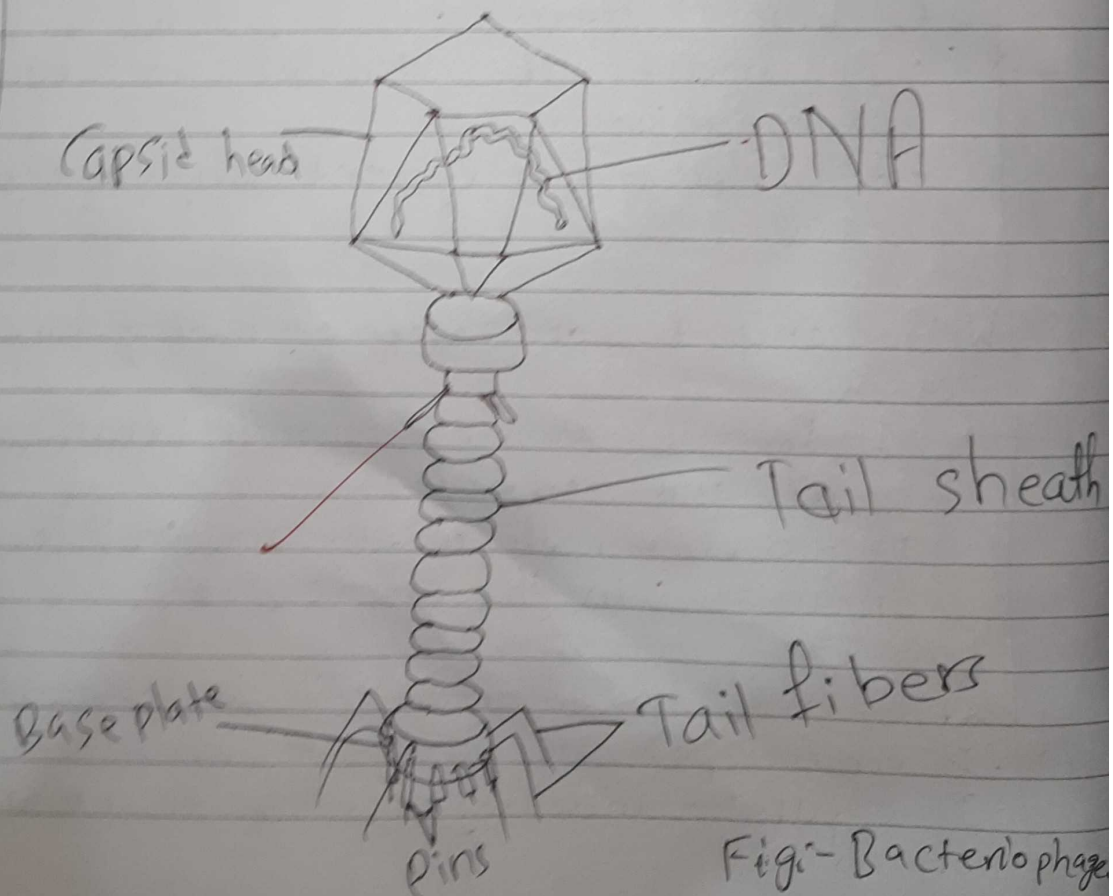


Fig- Bacteriophage

8) Draw a structure of mushroom.

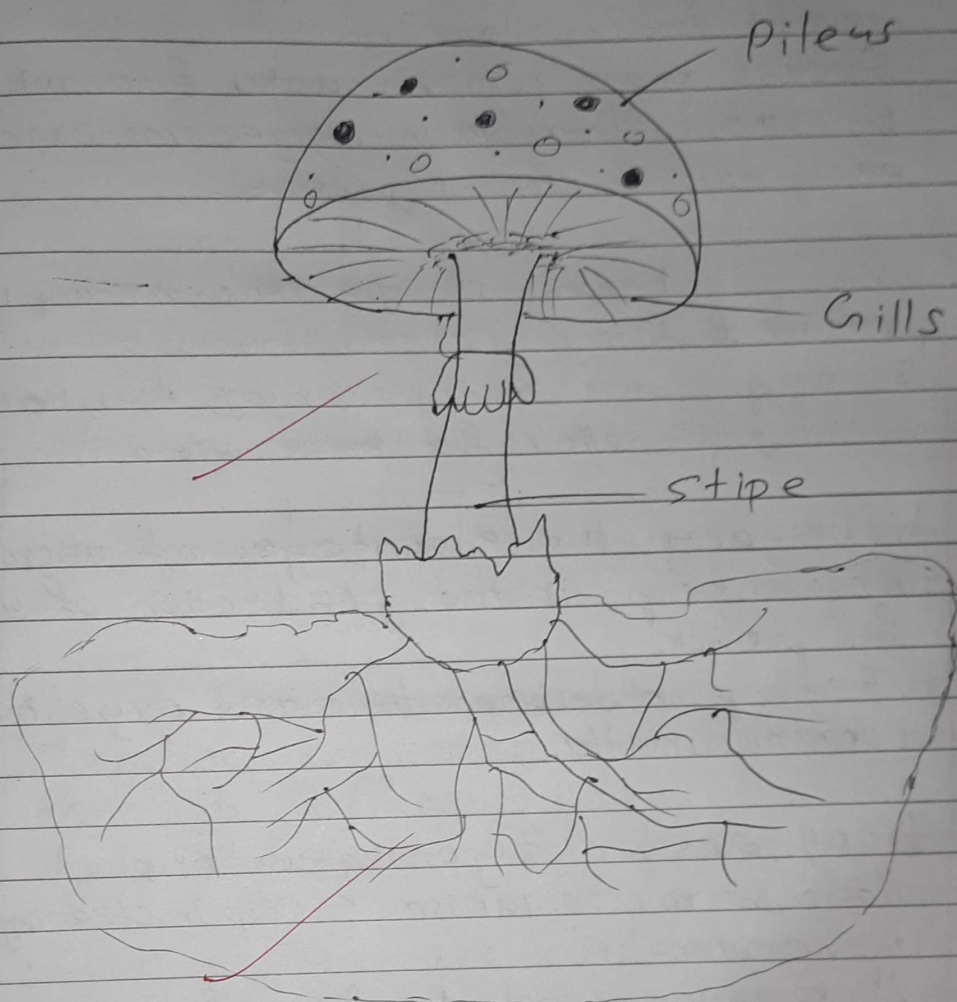


Fig:- Mushroom

9) Write the useful effects of fungi.

- Fungi are used in making bread and alcoholic beverages.
- Some fungi are used to produce antibiotics.
- Mushrooms are used as food.
- They decompose dead and decaying materials.

10) What is the role of bacteria in increasing soil fertility?

- Some bacteria, such as *Rhizobium*, fix atmospheric nitrogen and convert it into forms that plants can use. Thus, they increase soil fertility.

(1) How does addition of sugar in excess prevent food spoilage?

→ Excess sugar removes water from microorganisms by osmosis. Therefore, microorganisms cannot grow easily and the food is preserved.

(2) What will happen if you consume improperly preserved food?

→ It may cause food poisoning, diarrhoea, vomiting and other food-borne diseases.

(3) Write any two advantages of microorganisms.

- They help in the production of useful food products.
- They help in decomposition and recycling of organic matter.

(4) Long answer type questions.

1) How do microorganisms help in cleaning the environment?

→ Microorganisms help to clean the environment by decomposing dead plants, animals and other organic wastes. Bacteria and fungi convert these wastes into simpler substances. They also help in the decomposition of sewage and other biodegradable wastes. Thus microorganisms help in recycling nutrients and keeping the environment clean.

2) Describe the different methods of food preservation.

→ The common methods of food preservation are:-

- Drying - Removes moisture from food and prevents the growth of microorganisms.

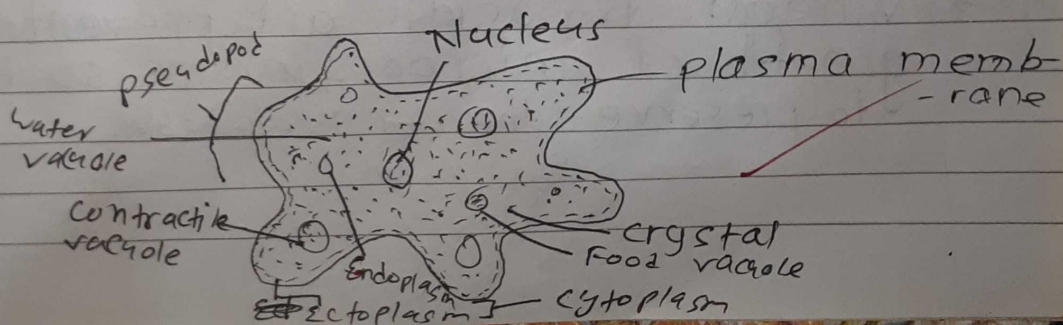
- Salting - Excess salt prevents microorganisms from growing.
- Adding sugar - Excess sugar removes water from microorganisms and prevents their growth.
- Pickling - Salt, oil and vinegar prevent the growth of microorganisms.
- Refrigeration - Low temperature slows down the growth of microorganisms.
- Heating - Heating kills many microorganisms present in food.
- Air-tight storage - Prevents contamination and growth of microorganisms.

3) How do microbes enter the human body?
→ Microbes can enter the human body through different ways:

- Food and water - By consuming contaminated foods or water.
- Skin - Through cuts and wounds.
- Insects - Through bites of mosquitoes, flies and other insects.
- Direct contact - Through contact with an infected person or contaminated objects.

4) Draw the neat and labelled diagram of amoeba.

→



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5) One day, Dipika saw her mother making pickle. Her mother cut the carrots turnips and cauliflowers into small pieces, washed them and put them in the sun for few hours. Then she mixed ~~common~~ common salts, oil, paste of onion, ginger and garlic vinegar etc as per requirement with the cut vegetables and heated them. After cooling she put the contents in air tight jar and kept it in the sun for many days.

Study the above paragraph and answer the following questions.

- a) Why did Dipika's mother cut the vegetables into small pieces and put them in the sun for few hours?
- She cut the vegetables into small pieces and dried them in the sun to remove moisture and prevent the growth of microorganisms.
- b) Why did she mix common salt in the cut vegetables and heated it?
- Salt helps to prevent the growth of microorganisms, while heating kills many microorganisms.
- c) Why did she mix vinegar?
- Vinegar is acidic and prevents the growth of microorganisms so it helps to preserve the vegetables.

Write the advantages of this process.

- It prevents food spoilage.
- It increases the shelf life of vegetables.
- It reduces the growth of microorganisms.
- The vegetables can be stored for a long time.

Study the diagram given alongside and answer the following questions.

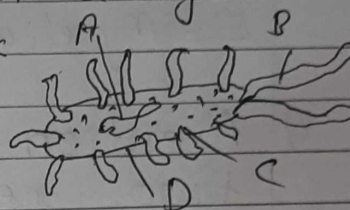
a) Label the parts A, B, C and D.

→ A - Genetic material (DNA)

B - Flagellum

C - Cell wall

D - Pili



b) Mention the kingdom with two reasons in support of your answer.

→ Kingdom - Monera

* Reasons - It is a unicellular prokaryotic organism.

- It has no true nucleus.

c) What is the mode of nutrition occur in the organism?

→ Bacteria show autotrophic as well as heterotrophic nutrition.

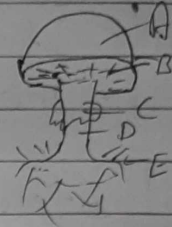
d) Write the method of reproduction occur in the organism.

→ Bacteria mainly reproduce by binary fission.

7) Study the diagram given alongside and answer the following questions.

a) Label the parts A, B, C, and D.

- A - Cap
B - Gills
C - Ring
D - stalk
E - Mycelium



b) Mention the kingdom with the two reasons in support of your answer.

→ Kingdom - Fungi

* Reasons - It does not contain chlorophyll, so it cannot prepare its own food.

- Its body is made up of thread-like structures called hyphae.

c) What is the mode of nutrition? ^{occur in the} ~~Organism~~

→ Fungi show saprophytic nutrition. They obtain food from dead and decaying organic matter.

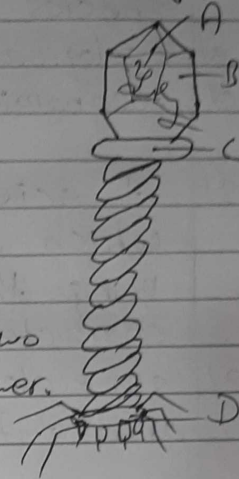
d) How does it help to control environmental pollution.

→ Mushrooms and other fungi decompose dead and decaying organic matter. Therefore, they help in decomposing wastes and recycling nutrients, which reduces environmental pollution.

8) Study the diagram given alongside and answer the following questions.

a) Label the parts A, B, C and D.

- A - Genetic material
B - Head
C - Tail sheath
D - Tail fibres



b) Mention the kingdom with two reasons in support of your answer.

→ /

c) Write any two nonliving natures of it.

- • It can be crystallized like non-living matter.
• It cannot reproduce independently outside a living host cell.

d) Write its two importance.

- • Bacteriophages can be used to destroy harmful bacteria.
• Viruses are useful in scientific and medical research.

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* Unit - 3.2 *

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